Generating Offline Signatures with a Variational Autoencoder and WGAN-GP with Feature Matching

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Abstract—We implement two generative models for offline signatures: a Variational Autoencoder (VAE) for reconstruction and latent sampling, and a Wasserstein GAN with Gradient Penalty (WGAN-GP) augmented with feature matching for higher-quality synthesis. Training uses a Google-Drive ZIP copy of a signature dataset with light augmentation. We report loss curves, reconstructions, and sample grids, and we summarize key hyperparameters and a VAE reconstruction score.

Index Terms— Variational Autoencoder, WGAN-GP, Feature Matching, Signatures, Generative Models

I. INTRODUCTION

Handwritten signatures present subtle stroke patterns and limited data, making them a good test-bed for generative modeling. VAEs encourage smooth latent spaces but may blur details; GANs aim for sharper realism but require stabilization. We implement both and compare qualitative behavior.

II. DATASET AND PREPROCESSING

Dataset is the signatures set from Assignment #1, provided as a Drive ZIP. Preprocessing converts to grayscale float tensors in [0,1], resizes to a fixed canvas, and applies mild augmentation (rotation, scaling, jitter, Gaussian noise) to improve generalization.

III. MODELS

VAE. Encoder → μ, $\log\sigma^2$; reparameterization $z = \mu + \sigma - \varepsilon$; decoder reconstructs x-. Objective is BCE(x, x-) + β·KL(q(z|x)-N(0,I)); β warmed up to 1.0. Sampling from N(0,I) yields new signatures.

WGAN-GP + Feature Matching. Critic maximizes E[D(real)] − E[D(fake)] with gradient penalty; generator minimizes −E[D(fake)] plus an L2 feature-matching term that aligns intermediate critic activations on real/fake batches. TTUR uses separate learning rates for G and D; n_critic=5.

IV. TRAINING SETUP

VAE uses Adam (lr≈2e-4) for 40 epochs with β-warmup.

WGAN-GP trains for 80 epochs with λgp=10, n_critic=5, TTUR (e.g., 1e-4 / 1.5e-4 for G/D), and feature-matching weight≈10. We log per-epoch losses and save sample grids.

Item	Value
Run timestamp	2025-11-02 07:21:28.530104
Train images	1649
Test images	500
VAE epochs / lr / β_final / wa	mup40 / 0.0002 / 1.0 / 10
WGAN-GP epochs / n_critic /	λgp / TTUR G/D / FM-W80 / 5 / 10.0 / 0.0001/0.00015 / 10.0
Test reconstruction (BCE) ↓	0.259494

Table I. Run summary and key hyperparameters parsed from logs.

V. RESULTS

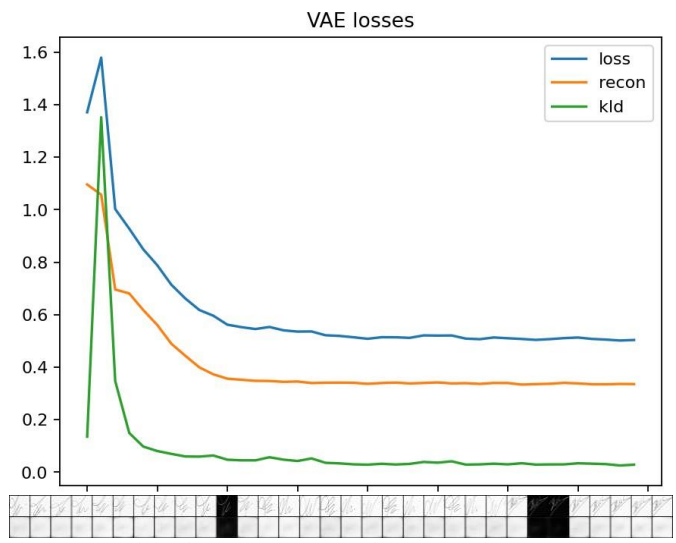
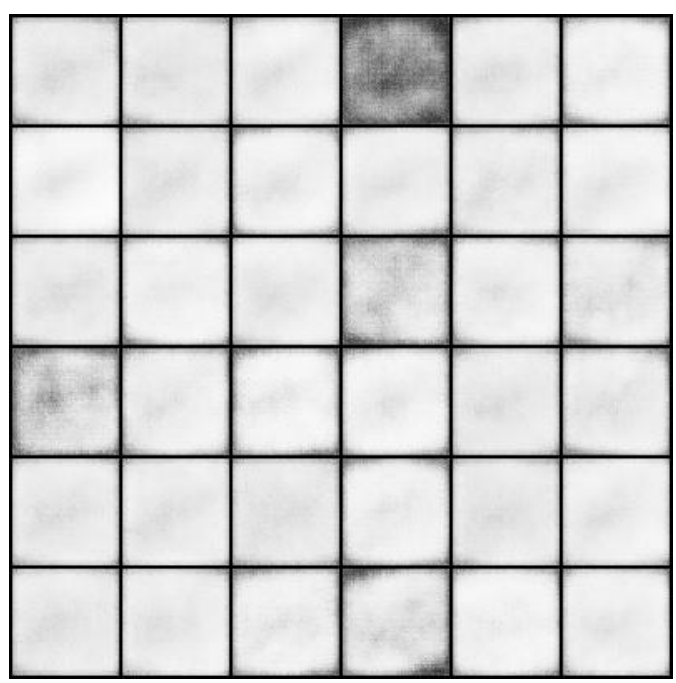


Fig. 1. VAE training loss components (total, reconstruction, KL).
Fig. 2. VAE reconstructions on test images.



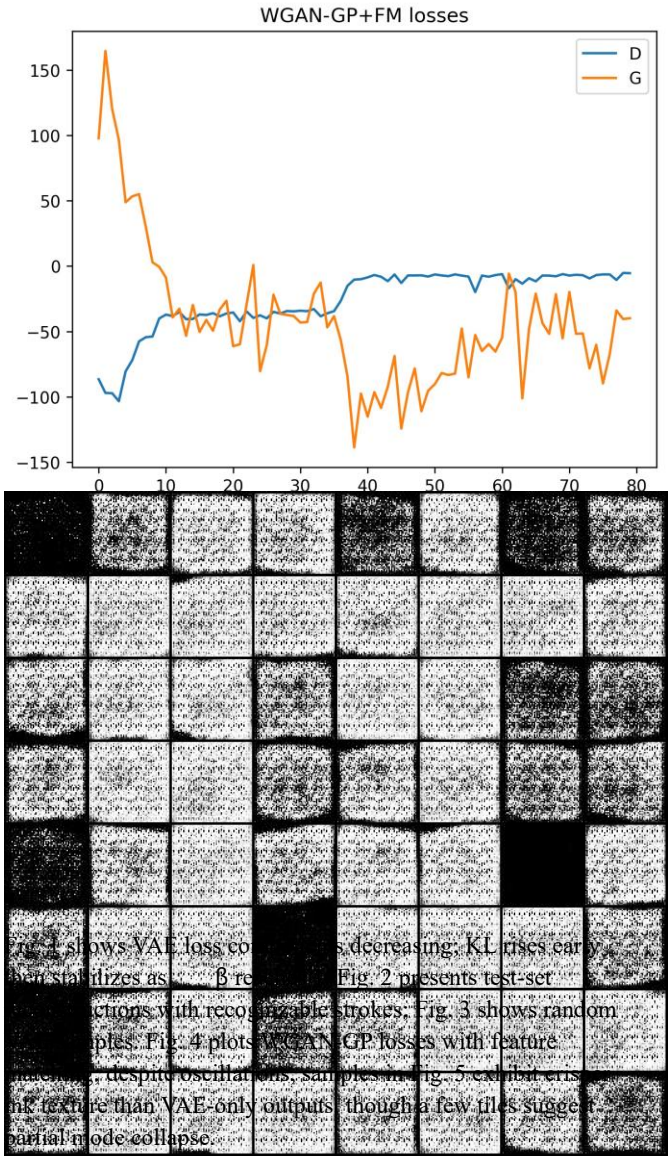


Fig. 5. Final WGAN-GP+FM sample grid.

VI. DISCUSSION

VAE provides stable training and a navigable latent space but smooths thin strokes. WGAN-GP improves sharpness; feature matching moderates divergence. To further improve realism, we could add spectral normalization, data-balanced sampling, and a perceptual or MS-SSIM term for reconstruction.

VII. CONCLUSION

On limited signature data, VAE yields coherent reconstructions and WGAN-GP adds sharper synthesis. Combining perceptual losses with stronger critic regularization is a promising next step.

VIII. PROMPTS

Prompts used: mount Drive + unzip; index images; grayscale/resize/augment; train VAE (β-warmup); train WGAN-GP

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